ORIGINAL ARTICLE

Unsupervised biomarker discovery leveraging foundation models: a multimodal approach to clinical data integration

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Abstract

In recent years Vision language models (VLM) have significantly gained popularity to process and analyze medical imaging data. The features extracted by vision encoders from the VLMs can be used in different activities like embedding with relevant text prompts or forming unsupervised clusters to find hidden biomarkers. The major challenge in digital pathology is that a huge portion of available datasets are unlabeled or partially labelled which makes the segmentation or classification tasks difficult. Unsupervised learning is used to find hidden patterns from the unlabeled Whole Slide Images (WSIs) by extracting image embeddings and by finding significant features as potential biomarkers. In this study we introduce "Unsupervised VLM-Driven Biomarker Identification and Meta-Integration (UVLMBI)," a novel framework that is designed to leverage VLM capabilities in unsupervised biomarker discovery and integration against relevant clinical metadata. Our method is used to extract rich feature embeddings from whole slide images of LungMAP dataset, devoid of explicit labels, with an available sparse metadata. We applied a suite of state-of-the-art VLMs, with most of them pre-trained on pathology-related image datasets (UNI, Prov-Gigapath, Conch, and Plip), while some of the VLMs were used as generalized foundation model applicable to medical imaging data (such as Llava and Llava-next). Our approach begins with extraction of image embeddings from the image encoders of each VLMs. Subsequently multiple unsupervised clustering algorithms, such as K-Means, Agglomerative, Gaussian Mixture, and Spectral are applied on the embeddings. Based on the clustering performances, we identify the optimal clustering – VLM combinations for the later experiments. One key novelty of the UVLMBI is the integration of clinical metadata to enhance the biomarker discovery. We develop predictive models to identify relevant biomarkers by linking image-derived clusters to clinical metadata based on patient demographics, clinical history, and molecular data. Beyond the mere identification of putative biomarkers, the integration delivers information about their clinical relevance in the light of personalized medicine strategies. Our experimental results show the effectiveness of UVLMBI framework in finding clinically significant biomarkers and provide comprehensive evaluations for collections of VLMs in unsupervised medical image analysis. This framework opens new dimensions of advanced data-driven techniques in digital pathology and opens new ways of research and clinical applications. This study provides an exploratory investigation of using VLMs for unsupervised biomarker discovery, acknowledging the challenges in clustering performance with complex medical image datasets. Our experimental results on a set of WSIs from human developing lung tissues in

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combination with clinical metadata shows promising results and can lead to unsupervised biomarker discovery in computational pathology.

Keywords Vision language model · Model fusion · Biomarker discovery · Unsupervised clustering · Multimodal AI · Lunge diseases

1 Introduction

Integration of artificial intelligence (AI) into digital pathology has been going through a steep growth, particularly deep learning (DL) models have been the most popular for solving computational pathology image analysis problems. The early works reported in the literature utilized CNNs for various tasks like tumor detection and segmentation on histopathological images. While effective, these methods require large, labeled datasets, which are often unavailable in the case of medical imaging. Vision language models (VLMs) have emerged as powerful tools for processing and analyzing medical images. VLMs like contrastive language-image pretraining (CLIP) [1], a large-scale image and noisy text embedding (ALIGN) [2], and others were initially developed for general image-text alignment tasks, but their application in medical imaging has shown great promise. These models use large-scale pre-training of image-text pairs and learn rich visual representations that can then be fine-tuned or directly applied to a variety of specific medical imaging tasks. Works like Radford et al. [1], and Jia et al. [2] created the foundation for such models in wide areas, including medicine. Unsupervised learning techniques in digital pathology are particularly valuable due to lack of labeled data. Clustering algorithms, such as K-Means, Agglomerative, Gaussian Mixture, Spectral, etc., are used to discover effective patterns from the unlabeled WSI data. These methods are used to identify tissue types or pathological features without requiring manual annotations. For example, Gaussian Mixture is effective in detecting morphological patterns in WSIs that are predictive of patient outcomes [3] Spectral clustering is good for non-linearly separable data, which is ideal for identifying complex histological patterns [4]. Agglomerative clustering can create hierarchical structures, which is valuable in capturing nested structures within histopathology images [5]. K-Means clustering is popular due to its simplicity and efficiency, especially in segmenting tissue types by grouping similar pixel intensities [6]. Fusing embeddings from multiple models to enhance the performance has been explored in various domains, including medical imaging.

Fusion methods combine embeddings from different sources to represent a more comprehensive format of data. In VLMs, different studies show improvement of model performances by fusing image-text information. Yang el al [7] and Dai et al. [8] proved the effectiveness of fusion in improving model robustness and accuracy. Huo et al. [9] showed the fusion of global and locally captured features improve image classification accuracies. Integration of clinical metadata with image features has become a growing area of interest in digital pathology. Patient metadata, such as demographics, clinical history, genomic records etc., can provide additional interpretability and clinical relevance of the identified biomarkers. Steyaert et al. [10] showed the integration of multimodal patient dataset can make the model performance more robust in finding incongruity of complex diseases to enhance the scope for personalized medicines. Study by Stahlschmidt et al. [11] showed the integration of clinical data with omics or image data features can help in finding out more complex patterns than the unimodal datasets. Braman et al. [12] showed that incorporation of clinical data along with other modalities, such as radiology, histopathology and omics, outperformed the state-of-the-art unimodal model.

Vision language models offer distinct advantages over traditional unimodal approaches by leveraging both image and textual information in identification of biomarkers. VLMs, pretrained with diversified datasets, are capable of extracting complex and multi-dimensional features from WSIs that might be missed by conventional methods pretrained with either image or textual data alone. However, most of the recent studies focus on either single VLMs or clustering methods without considering the synergistic effects of combining multiple VLM outputs and clustering algorithms. This gap is highly important, since no single VLM or clustering method can evoke

all the nuances within WSIs, especially when combined with clinical metadata. In the current study we attempted to fill this deficit by proposing the UVLMBI framework that not only integrates multiple VLMs but also applies advanced clustering techniques and fuses the VLM outputs using methods such as stacked autoencoders. This approach tries to provide an insight into the pathological processes that could have been missed out due to a more isolated approach. However, complete optimization and validation for such complex integrations across a wide spectrum of pathological conditions remain open and a challenge for further research.

The entry of artificial intelligence has ensured outstanding improvements in the field of digital pathology, especially with the incorporation of Vision-Language Models (VLMs). These models, developed originally for the processing and understanding of visual content and its corresponding textual description, are finding growing applications in the domain of medical imaging, where they hold enormous potential for the analysis of Whole Slide Images (WSIs). VLMs scan and digitize microscopic slides at high resolution, with deep substantive information that mostly goes underutilized, partly because of challenges with manual annotation and interpretation of WSI data. In this regard, the use of VLMs creates room for automatic extraction of meaningful features in these images that can henceforth be used in several downstream tasks, including biomarker discoveries. One of the major difficulties in digital pathology is the inadequate large datasets with proper labeling which makes supervised learning techniques difficult to apply. In this context, unsupervised learning proves to be a useful tool, having the potential to introduce hidden patterns or structures within the data as a discovery of new biomarkers that indicate any biological state or condition. Biomarkers detected in the context of medical imaging are of utmost importance in diagnosing diseases, predicting patient outcomes, and recommending personalized treatment plans. Therefore, one of the most promising approaches toward overcoming the limitations imposed by unlabeled datasets in digital pathology involves the integration of VLMs with unsupervised learning methods.

This paper introduces the "Unsupervised VLM-Driven Biomarker Identification and Meta-Integration" (UVLMBI) framework, a detailed pipeline to effectively leverage the full potential of VLMs in biomarker exploration using unlabeled WSIs. Using the LungMAP dataset [13], a highly valuable resource of images in lung tissue paired with clinical metadata, the UVLMBI framework makes use of a range of pre-trained VLMs such as UNI [14], Prov-Gigapath [15], Conch [16], Llava [17], Llava-Next [18], and Plip [19] to obtain important image embeddings. These embeddings are then used for unsupervised clustering in next steps with K-Means, Agglomerative Clustering, Spectral Clustering, and the Gaussian Mixture Model. One critical achievement in the UVLMBI framework is that it fuses these diverse VLMs into a single unified representation using stacked autoencoder. This fusion process may therefore allow strengths of individual VLMs to create even more robust and comprehensive embeddings, potentially capturing a wider range of features that are potentially missed by any one model. The performance of these clustering algorithms is, therefore, investigated in this work with the use of individual and fused VLM embeddings to determine which combination works best in biomarker discovery. Another main feature of the UVLMBI framework is the integration of clinical metadata with the extracted image features in overall analysis. It consists of a set of metadata such as patient demographics, clinical history, and molecular data, and has been combined with the image-derived clusters in order to enhance the relevance and interpretability of the identified biomarkers. In this approach, those clusters linked with metadata attributes are designed to underline the biomarkers that are representatives of some biological processes and clinically relevant for the rationale of personalized medicine. This work represents a first step in leveraging VLMs for biomarker discovery in complex, high-dimensional medical image data. While the unsupervised clustering results face challenges in cluster separability, this is a common issue in medical image analysis, where heterogeneity and noise are prevalent. To the best of our knowledge, our work is the first one to try a fusion of multiple VLMs along with metadata integration and highlights the feasibility of integrating these models and paves the way for future refinements.

The main contributions from this study are:

- Introduction of the UVLMBI framework for unsupervised biomarker discovery using Vision-Language Models (VLMs) and clinical metadata integration.

- Evaluation of state-of-the-art VLMs (UNI [14], Prov-Gigapath [15], Conch [16], Llava [17], Llava-Next [18], Plip [19]) and their fusion in medical imaging for biomarker identification.
- Application of advanced unsupervised clustering algorithms to identify optimal combinations of VLMs, including fused representations, and clustering methods.
- Integration of clinical metadata to enhance the interpretability and clinical relevance of identified biomarkers which can lead to personalized medicine by linking image-derived clusters with patient-specific clinical data.

The rest of the paper is organized as follows. Section 2 introduces the UVLMBI framework with fusion of VLMs. Section 3 provides detailed experimental results on biomarker prediction using the proposed approach. Finally, Sect. 4 concludes the paper with future directions.

2 Unsupervised VLM-driven biomarker identification framework

The comprehensive workflow for the unsupervised vision-language model-driven biomarker identification and meta-integration (UVLMBI) framework is shown in Fig. 1. It begins with processing the WSIs, using color normalization and patches creation. The 256 × 256 sized patches are generated from the WSIs, which are essential for the further downstream processes. Based on the age groups, these patches are input to the pre-defined VLMs

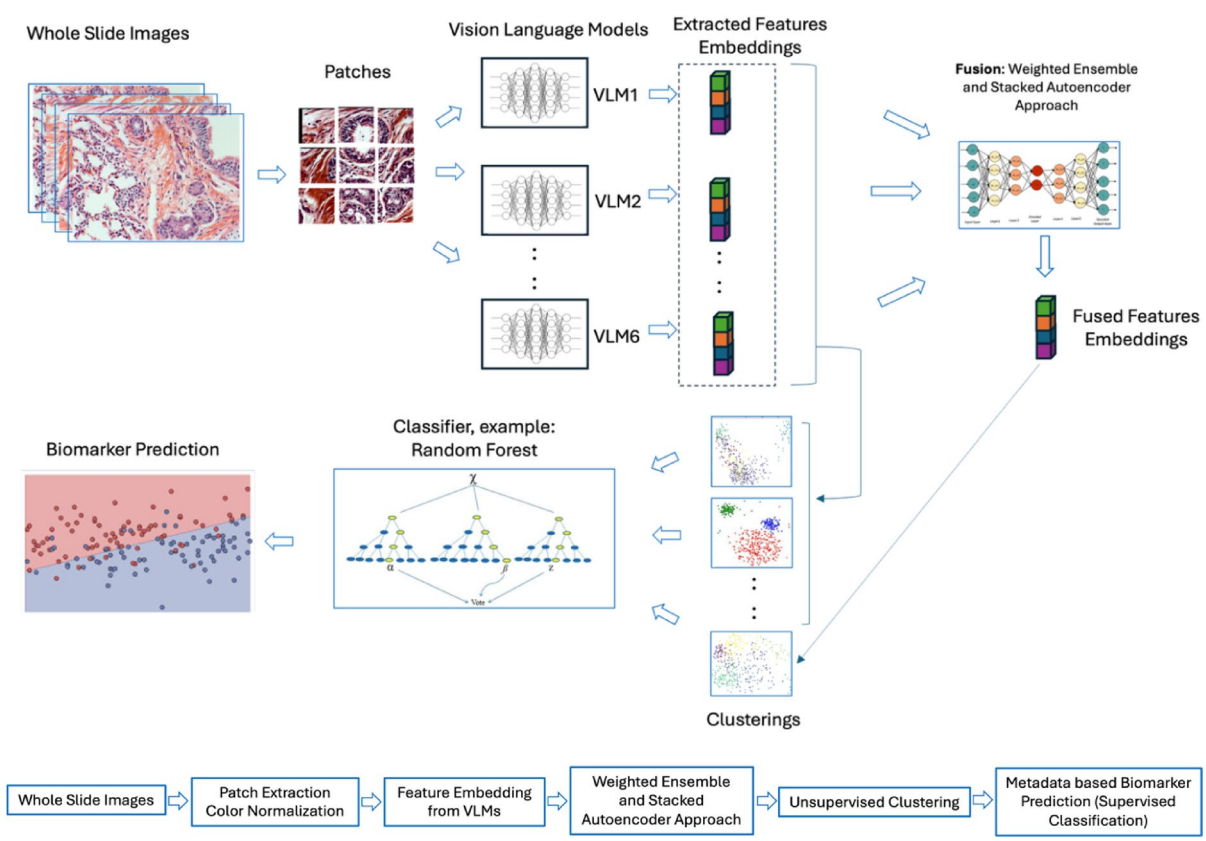

Fig. 1 Overview of the proposed UVLMBI framework for unsupervised biomarker discovery. The pipeline begins with whole slide images (WSIs) that undergo color normalization, followed by patches creation into smaller regions for processing. Features are extracted from each patch using multiple Vision-Language Models (VLMs). The extracted features undergo fusion and dimensionality reduction via a stacked autoencoder. Subsequently, unsupervised clustering identifies potential biomarker groups. Finally, these identified clusters are used in a supervised classification task, integrated with clinical metadata, for biomarker prediction and validation. Simplified flowchart illustrates the sequential steps of our methodology

to extract relevant feature embeddings which contain important morphological information. The extracted feature embeddings are fused using weighted ensemble method and a stacked autoencoder. The stacked autoencoder reduces the dimensionality and combines the strengths of each individual model into a unified representation. These extracted and fused embeddings are then subjected to different clustering algorithms to get potential clusters, which group similar patches and reveal potential biomarkers. Further, clinical metadata (e.g. patient demographics etc.) and the clustering results are integrated to predict the biomarkers using machine learning classifier, e.g. Random Forest. This prediction helps in identifying clusters that may be indicative of specific biological or histopathological conditions, biomarker discovery. The complete workflow of our approach is depicted in the simplified flowchart in Fig. 1 bottom, providing a clear overview of the sequential processing steps from raw images to biomarker prediction. In UVLMBI, clustering provides label-free 'biomarker groups' in the image-feature space. The supervised stage is not a direct image classifier here. Instead, we use patient metadata to predict the discovered cluster labels at the slide/patient level, quantifying how well clinical context aligns with the unsupervised image-space structure.

2.1 Dataset and preprocessing

The LungMAP dataset (https://www.lungmap.net), which forms the basis of our study, consists of thousands of whole slide images stained by three different techniques: Hematoxylin and Eosin (H & E), immunofluorescence-confocal, and in situ hybridization (ISH). These samples represent both human and mouse tissues and provide a rich resource in examining lung tissues across a range of conditions and ages. The LungMAP project provides valuable histopathology data focused on lung development and diseases [13]. These diseases are characterized by distinct tissue remodeling, inflammation, and fibrosis, which are captured in histopathology slides to help researchers identify key biomarkers and understand the underlying pathological mechanisms. LungMAP's large dataset of high-resolution tissue images enables detailed analysis of disease progression and potential therapeutic targets in lung pathologies.

In this paper, we are particularly interested in human tissues stained with H & E. We note that the H & E-stained pathology imaging is the most common and widely used modality that captures nuclei-cytoplasmic contrast essential for morphology-based biomarker discovery. Moreover, several of the VLMs we evaluate are trained specifically on H & E histology. The samples were chosen by six different age groups, including 1 day, 6 days, 4 months, 9 months, 20 months, and 4 years. On average, 40 WSIs from each of the age groups were randomly chosen to ensure that the sample is representative and diverse. This sample size balances computational tractability with representation of inter-donor and intra-tissue variability. All the WSIs used in this paper were captured by using 20X magnification whose original resolution is 1230×899 pixels.

Histogram Normalization: We performed histogram normalization to rectify the variations in staining and illumination that might affect the analysis. It is a process of standardizing the intensity distribution in images, making them more comparable across various WSIs. It normalizes the contrast of the image, reshaping the distribution of pixel intensity to fit a target histogram shape, usually a uniform or Gaussian distribution. It reduces, in a very effective way, the impact of staining variation and any other artifacts on improving the robustness of subsequent image analysis tasks.

Mathematically, histogram normalization can be represented as:

$$I_{norm}\left(x,y\right) = {CDF_t}^{-1}(CDF_s\left(I\left(x,y\right)\right), \tag{1}$$

where *I(x,y)* is the original pixel intensity at position *(x,y)*, *CDFs* is the cumulative distribution function of the source image, *CDFt* is the cumulative distribution function of the target histogram.

Tiling: Normalization was followed by splitting the WSIs into smaller segments, or patches, of 256×256 pixels. This patches creation process facilitates the application of AI models and helps in focusing on the most relevant regions of tissue within every slide. Therefore, patches that contained mostly white space or negligible

amounts of tissue were excluded from analysis to make sure that only the informative regions were analyzed. The number of patches obtained for each age group was as follows: 1 day—122 patches; 6 days—62 patches; 4 months—450 patches; 9 months—449 patches; 20 months—122 patches; 4 years—462 patches. Although the number of extracted patches per age group varied, this is done to directly reflect tissue area differences among slides. We intentionally maintained all informative patches rather than enforcing equal patch counts, because down- or over-sampling could remove biologically relevant morphology or replicate redundant regions. The dataset is already balanced at the slide level (≈ 40 WSIs per age group), and normalization plus white-region filtering ensured comparable tissue coverage across groups.

The preprocessing workflow considers standardization and optimization for the successive analysis stages, merging histogram normalization with selective patches creation. Further, these patches were used as input by several Vision-Language Models (VLMs), where features are extracted to be used in later steps of this work regarding clustering and biomarker identification.

2.2 Feature extraction from vision-language models (VLMs)

Vision-Language Models (VLMs) have become powerful tools in medical imaging due to their ability in understanding both image and text data. The VLMs consist of image and text encoders. The image encoder extracts features from the images and text encoder processes accompanying texts. The two modalities are embedded to make sense of the image in the context of the text. In this project, we used six different state-of-the-art VLMs which are pretrained using histopathological image datasets. In this project, we used 6 state-of-the-art VLMs. While models like Conch, Prov-Gigapath, Plip and UNI are pretrained on histopathological datasets, Llava and Llava-Next, though trained on general image-text pairs, were included to evaluate their adaptability in extracting meaningful features from pathology images and to explore the potential of generalized foundation models in this domain.

2.2.1 Prov-Gigapath

Prov-Gigapath [15] is an advanced pathology foundation model designed to analyze gigapixel pathology slides. It leverages real-world clinical data from the Providence health network through the pretraining on 1.3 billion image tiles from over 170,000 whole-slide images, representing more than 30,000 patients and 31 tissue types. In this work, a new vision transformer architecture, GigaPath, combined with LongNet, in a way that efficiently handles the large-scale context of gigapixel slides. Prov-Gigapath is an advanced pathology-based foundation model designed to analyze gigapixel level pathology slides by leveraging a new vision transformer architecture called GigaPath. GigaPath is specifically engineered to handle the large-scale context of these immense slides. It achieves this by combining with LongNet [33], an architectural component that is crucial for processing the long sequences of image tiles that constitute a gigapixel slide. This combination allows the model to set a new performance standard on 25 out of 26 digital pathology tasks, including cancer subtyping and mutation prediction. It sets a new performance standard on 25 out of 26 digital pathology tasks, including cancer subtyping and mutation prediction, and significantly outperforms other models on many benchmarks.

2.2.2 Conch

Conch[16] is a vision language foundation model built for computational pathology. It is pre-trained over 1.17 million image-caption pairs from diverse sources, which is considered the largest histopathology-specific paired image-text dataset to date. Conch utilizes a contrastive learning approach to align visual and textual modalities, enabling it to perform a wide range of tasks, including zero-shot classification, segmentation, image captioning, and cross-modal retrieval. The model's architecture is based on CoCa (contrastive captioners) and incorporates an image encoder, a text encoder, and a multimodal fusion decoder. This model sets new benchmarks for

performance across 14 downstream tasks, outperforming previous models such as PLIP and BiomedCLIP [20] by 12% to 17% on key histopathological classification tasks, including lung adenocarcinoma subtyping and colorectal cancer tissue classification. Conch has proved to be a highly versatile model for a variety of pathology applications.

2.2.3 Llava

Llava [17] is a multimodal model designed to integrate both vision and language for a wide range of tasks, from visual question-answering to complex reasoning. It is built using a pre-trained CLIP visual encoder [1] (ViT-L/14) paired with the Vicuna language model [21], leveraging the strength of both visual and language-based learning. Llava is trained on 595K image-text pairs and fine-tuned on a 158K multimodal instruction following samples including tasks like conversations, image descriptions and complex reasoning. Llava achieved an 85.1% relative score compared to GPT-4 on multimodal instruction-following datasets. When fine-tuned on the ScienceQA dataset, Llava achieves a state-of-the-art accuracy of 92.53%, setting a new benchmark for multimodal reasoning tasks.

2.2.4 Llava-Next

Llava-Next [18] is an advanced multimodal model designed to enhance reasoning skills, OCR, and world knowledge capabilities. Built on the success of Llava-1.5, Llava-Next increases input image resolution by four times, supporting flexible resolutions of up to 1344×336. It improves visual conversation and reasoning across diverse applications. Trained on a dataset of 1.3 million samples, LLaVA-NeXT outperforms many open-source and commercial models, such as Gemini Pro [22] and Qwen-VL-Plus [23], in benchmarks for visual reasoning, OCR, and multimodal understanding. LLaVA-NeXT-34B achieves state-of-the-art performance in zero-shot scenarios maintaining efficient deployment with minimal training costs.

2.2.5 Plip (pathology language-image pretraining)

PLIP [19] is a multimodal AI model trained with a large-scale dataset from OpenPath pathology images and their corresponding natural language descriptions. The dataset contains over 208,000 pairs of pathology images and texts. By utilizing contrastive learning, PLIP attains state-of-the-art zero-shot classification performance on various pathology datasets and classifies new pathology images without the need for retraining. It is superior compared to the existing models, such as CLIP, in classifying, retrieving, and understanding pathology images, and it leverages publicly available medical Twitter data to easily enable the sharing of knowledge in diagnosis.

2.2.6 UNI

UNI [14] is a state-of-the-art, self-supervised vision model developed for computational pathology. Pretrained on a massive dataset of more than 100 million images sourced from over 100,000 diagnostic whole-slide images (WSIs) across 20 major tissue types, it represents one of the largest pretraining efforts in pathology. Using the DINOv2 self-supervised learning method, UNI can generalize across a wide range of clinical tasks, such as tissue classification, cancer subtyping, and biomarker screening. Evaluated on 34 distinct clinical tasks, including rare cancer classification and slide-level weakly supervised learning, UNI consistently outperforms existing models like CTransPath [24] and REMEDIS [25]. Its robustness in image retrieval, slide classification, and segmentation tasks, especially for rare and underrepresented diseases, makes UNI a rich foundational model for pathology AI, with the potential of large-scale, self-supervised learning in clinical workflows.

Each VLM processes an image patch X_i through its visual encoder β, resulting in a feature embedding F_i,

$$F_i = \beta\left(X_i\right). \tag{2}$$

These embeddings, typically high-dimensional vectors, encapsulate critical visual features such as texture, morphology, and spatial relationships within the tissue, which are essential for identifying pathological regions.

Figure 2 highlights the key step of weighted ensemble fusion and stacked autoencoders, where different VLMs extracted features are combined. The stacked autoencoder compresses the high-dimensional fused embeddings into a 64-dimensional latent space. This non-linear fusion process ensures that the combined features are more informative and ready for downstream tasks, such as clustering and classification. The embeddings are processed for each combination (age group—clustering algorithm pair), resulting in a fused embedding for each scenario.

2.3 Clustering of feature embeddings

We apply four different clustering algorithms – K-Means, Agglomerative, Gaussian Mixture, and Spectral – to the extracted VLM embeddings to identify potential biomarker clusters. These algorithms operate by grouping embeddings $\{F_1, F_2, \ldots, F_n\}$ into clusters $\{L_1, L_2, \ldots, L_n\}$,

$$L_i = Cluster(F_i) \tag{3}$$

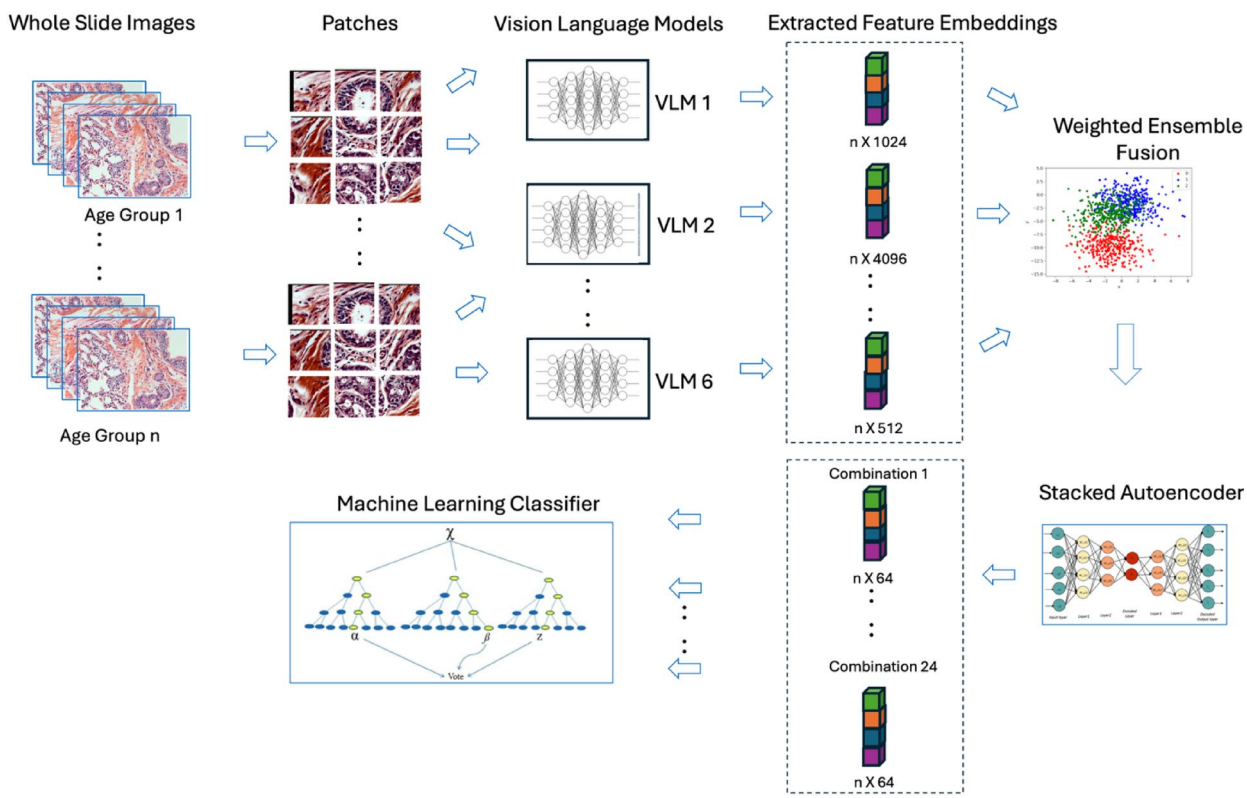

Fig. 2 Workflow illustrating the fusion and refinement of feature embeddings for biomarker prediction. Whole slide images (WSIs) from various age groups are color normalized and divided into patches. Features are extracted from each patch using multiple Vision-Language Models (VLM1 to VLM6), with each VLM producing embeddings of different dimensions (e.g., n × 1024, n × 4096, n × 512 etc. based on different VLMs). These features undergo weighted ensemble fusion based on silhouette scores, followed by dimensionality reduction and refinement via a stacked autoencoder, compressing them to n × 64 dimensions. The scatter plot shown for weighted ensemble fusion is a conceptual representation of the fusion process and not an actual data plot. Weighted fusion is performed before clustering, and no clustering is applied inside the fusion block. The resulting fused embeddings are then used for clustering and classification in various age groups and clustering algorithm combinations

Each cluster label L_i corresponds to a potential biomarker, representing a group of similar image patches.

2.3.1 K-Means clustering

This is a simple clustering method where the data is divided into a predefined number of clusters by minimizing the distance between data points and their corresponding cluster centroids. It iteratively assigns points to the nearest centroid, recalculates the centroids, and repeats the process until convergence. Though it is efficient and scalable, K-Means can be sensitive to the initial placement of centroids.

2.3.2 Agglomerative clustering

Agglomerative Clustering is a hierarchical method that starts by treating each data point as its own cluster and iteratively merges the closest clusters. This bottom-up approach does not require the number of clusters to be predefined, making it flexible for hierarchical data structures. However, its computational cost increases with larger datasets due to the repeated merging process.

2.3.3 Gaussian mixture model (GMM)

This method assumes data points are generated from a mixture of several Gaussian distributions. Unlike K-Means, each point has a probability of belonging to multiple clusters. This makes GMM more flexible for clusters of varying shapes and densities, though it can be computationally intensive for large datasets.

2.3.4 Spectral clustering

It is a graph-based method that uses the eigenvalues of a similarity matrix to reduce the dimensionality of the data before applying clustering. This approach is effective for datasets with complex, non-linear cluster boundaries. It performs particularly well when clusters have irregular shapes but can be slower to compute for large datasets.

2.4 Fusion of VLM embeddings

To exploit the strengths of various VLMs, we performed fusion using a stacked autoencoder. This type of fusion aims at forming a unified representation by combining the embeddings from different VLMs. The architecture of the stacked autoencoder consists of an encoder. The encoder ε compresses the concatenated embeddings from multiple VLMs into a lower-dimensional representation Z. The decoder D reconstructs the original embeddings from Z, ensuring the preservation of essential features.

The fusion process is mathematically represented as,

$$Z = \varepsilon(Concat(F^{(1)}, F^{(2)}, \ldots, F^{(m)})) \tag{4}$$

$$F = D\,(Z)\,, \tag{5}$$

where $F^{(1)}$, $F^{(2)}$,…, $F^{(m)}$ are embeddings from different VLMs, and Z is the compressed, fused representation.

2.5 Stacked autoencoder

A stacked autoencoder is a deep learning architecture composed of multiple layers of autoencoders, stacked together to form a hierarchical model. The encoder part compresses the input data into a lower dimensional representation, and the decoder part reconstructs the input form from its compressed representation. Let input from each VLM be denoted as X_i, where $i = 1, 2, \ldots, 6$ represents the six VLMs. The stacked autoencoder performs

encoding in multiple layers, compressing the inputs to a shared latent representation Z. The encoder for each layer can be described as,

$$Z^l = f\left(W^l\right) \bullet Z^{l-1} + b^{(l)} \tag{6}$$

where,

- $Z^{ll)}$ is the latent representation at the l-th layer,
- $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the l-th layer,
- f is a non-linear activation function,
- $Z^{(0)} = [X_1, X_2, ..., X_6]$, the concatenation of features from all six VLMs.

The decoder reconstructs the inputs as,

$$\widehat{Xi} = \mathrm{f}\ (\mathrm{W}'(\mathrm{l}) \,.\, \mathrm{Z}(\mathrm{l}) + \mathrm{b}'(\mathrm{l})) \tag{7}$$

where $W'^{(l)}$ and $b'^{(l)}$ are the decoding weights and biases.

The goal of training is to minimize the reconstruction loss $L = \sum_i \left\| X_i - \hat{X}i \right\|^2$, ensuring that the fused representation captures the relevant features of all six VLMs effectively.

2.6 Biomarker identification

A biomarker is defined as a cluster of patches with common visual features that could point to a specific biological or pathological state. Clusters that were stable across the VLMs and clustering algorithms were prioritized as the biomarker candidates. In this context, a 'configuration' refers to a unique combination of a VLM, an age group, and a clustering algorithm (e.g., Prov-Gigapath, 4 months, K-Means). We measure 'cluster stability' to quantify how consistently a specific cluster appears across these different configurations. Stability of a cluster C across different configurations is measured by,

$$S(C) = \frac{\sum_{i=1}^{k} I(C, L_i)}{k} \tag{8}$$

where $I(C,L_i)$ indicate the presence of cluster C in the i-th configuration, and k is the total number of configurations. Note that the configuration can point to a unique combination of age group, clustering algorithm, and embedding source (each VLM or the fused representation). Thus, $S(C) \in [0,1]$ measures the recurrence of C across settings.

2.7 Integration of clinical metadata

Clinical metadata were integrated to further enhance the clinical relevance of the identified biomarkers. Clinical metadata, including patient demographics, clinical history, and molecular data, were combined with the image-derived clusters for supervised learning tasks. The labels for this supervised learning task are the cluster assignments generated by the unsupervised clustering of the image patches. For a given patient, the classification label is determined by the most frequent or dominant cluster among all the patches from that patient's slide. This process allows us to predict the identified biomarker clusters from clinical metadata and associate these clusters with clinically meaningful outcomes. These predictions are done based on different machine learning classifiers, such as, Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). Classifiers are trained based on a combined feature set of image embeddings and metadata, while the performance is measured based on F1-scores.

The prediction task can be described as,

$$\widehat{y} = \text{Classifier (Xmetadata)} \tag{9}$$

where $X_{metadata}$ denotes the clinical metadata, and $\widehat{y}$ is the predicted cluster label. Clustering is performed at the patch level. For patient-level use, we aggregate patch assignments per patient using majority vote where ties are resolved by the cluster with the higher mean silhouette for that patient's patches.

Figure 3 shows the workflow for biomarker discovery and prediction using feature embeddings from the previously defined VLMs (Conch, Prov-Gigapath, UNI etc.) and their fused embeddings across 24 combinations of age groups and cluster algorithms. For each VLM and age group combination, features are extracted, clustered, and identified as unsupervised biomarker groups. These clusters represent potential biomarkers derived from the feature space. The clusters are then combined with patient specific metadata to form a dataset for the biomarker prediction. The dataset is processed through a machine learning classifier, for example Random Forest, to predict the biomarker labels based on the extracted clusters and metadata. The prediction output highlights the best-performing models across different age groups and clustering algorithms, helping to identify the most relevant biomarkers for each scenario.

2.8 Evaluation metrics

We applied Silhouette Score and Receiver Operating Characteristic Area Under the Curve (ROC-AUC) scores to measure the effectiveness of clustering and biomarker prediction.

2.8.1 Silhouette score

The silhouette score is a metric used to evaluate the consistency of data within clusters. It evaluates both intra-cluster cohesion and inter-cluster separation, making it suitable for assessing clustering quality without requiring any ground truth labels. While other metrics like Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) are also used in clustering tasks, they are more commonly applied when comparing clustering results to an external reference (i.e., ground truth labels). In contrast, Silhouette Score provides a direct assessment of the

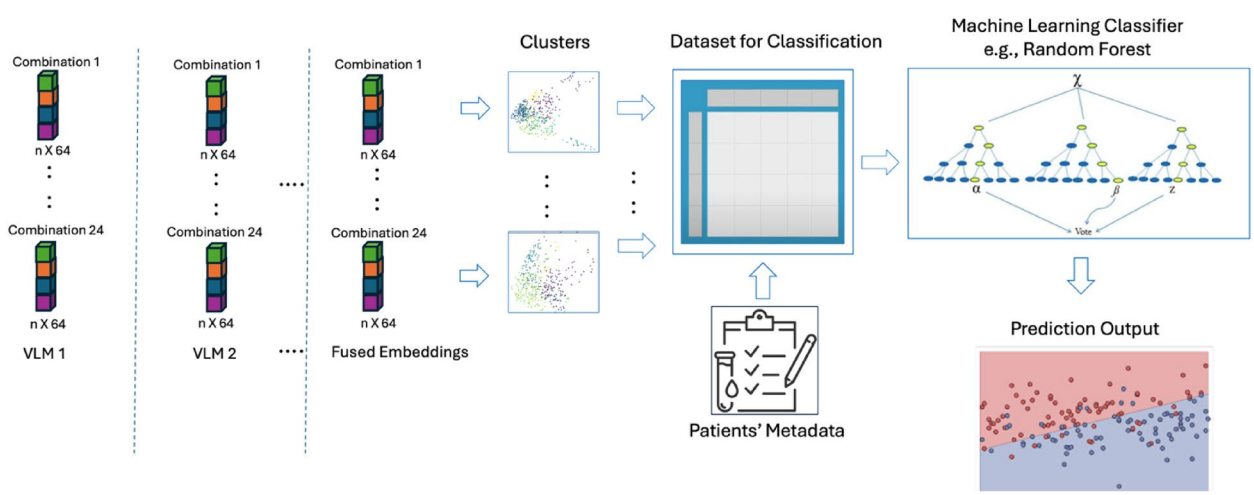

Fig. 3 Overview of biomarker prediction using extracted features from multiple Vision-Language Models (VLM1, VLM2, …) and fused embeddings for all age group-clustering algorithm combinations (24 in total). Features from each combination (age group × clustering algorithm) are extracted from VLMs and fused embeddings, then clustered to identify potential biomarker groups. These clusters, combined with patient metadata, form a dataset used to predict biomarkers through machine learning classifiers, such as Random Forest. The prediction output displays the best-performing model for each combination, revealing the most relevant biomarkers

structure within the data itself, which is important for truly unsupervised settings like here biomarker discovery, where no predefined classes exist. For each sample *í*, the silhouette score *s(í)* is defined as,

$$s(i) = \frac{\mathrm{b}(i) - \mathrm{a}(i)}{\max(\mathrm{a}(i),\ \mathrm{b}(i))} \tag{10}$$

where a(*í*) is the average distance between iii and all other points in the same cluster. b(*í*) is the minimum average distance between *í* and all points in the nearest cluster (not including *í*'s own cluster). The silhouette score ranges from −1 to 1, where a high score indicates that the sample is well matched to its own cluster and poorly matched to neighboring clusters.

2.8.2 Receiver operating characteristic—area under the curve (ROC-AUC)

The ROC-AUC for a model is calculated as,

$$\mathrm{ROC-AUC\ score} = \frac{\sum_{i=1}^{N} \mathrm{True\ Positive\ Rate} - \mathrm{False\ Positive\ Rate}}{N} \tag{11}$$

where *N* is the number of thresholds used to compute the ROC curve.

2.9 Visualization and interpretation

To compare the performances of different VLMs, clustering algorithms, and predictive models, we created cluster plots, ROC curves, and box plots. The metrics were also visualized to compare the performance of different models and algorithms across various age groups and clustering methods, with a particular focus on the ROC-AUC as a key indicator of model effectiveness.

3 Experimental results and discussion

This section outlines the experiments conducted to validate the proposed UVLMBI framework on the LungMAP dataset. We investigate the effectiveness of Vision language models in extracting useful features, unsupervised clustering for biomarkers discovery, and predictive modeling using clinical metadata. A combination of six pre-trained VLMs and a fused ensemble approach was evaluated for biomarker prediction across various clustering algorithms.

3.1 Dataset and preprocessing

The LungMAP dataset is used as the foundation of all experiments. It consists of Whole Slide Images (WSIs) from lung tissue samples, stained with Hematoxylin and Eosin (H & E) techniques. As mentioned before, we selected six age groups—1_day, 6_days, 4_months, 9_months, 20_months, and 4_years. From each age group we selected 40 WSIs randomly. After normalization and patches creation, each WSI is divided into patches of resolution 256 × 256 pixels. Then the patches with mostly white regions, were removed as they would not contribute useful information of the tissue regions. The final dataset consisted of the following number of patches per age group: 1_day—122 patches, 6_days—62 patches, 4_months—450 patches, 9_months—449 patches, 20_months—122 patches, and 4_years—462 patches. These patch counts, which are determined by the amount of viable tissue remaining after preprocessing, naturally vary across different age groups and biological stages. For each age group, these patches were processed with six different VLMs to extract feature embeddings. The codes were run in an environment setup with Nvidia A100 and V100 GPU clusters with 128 GB RAM.

3.2 VLM-based embedding extraction

The image encoders of the VLMs extract essential features out of the input patches. We utilize zero-shot inference with these VLMs, as four of them are pre-trained on histopathological H&E slides, while Llava and Llava-Next are large, generalized foundation models which were included to evaluate their adaptability in assessing pathological images to explore the potential of generalized foundation models in this domain. However, the extracted feature dimensions vary based on different VLMs. To fuse these features, we need to modify them first to a standard dimension. For this, we apply the Principal Component Analysis (PCA) technique that reduces and aligns the features for consistent fusion. We chose PCA for its computational efficiency and effectiveness in linear dimensionality reduction. Extracted features from the images are stored as list of features in.npy files based on the age groups. Table 1 summarizes the dimensions of the extracted features saved for each age group. Once fused, a stacked autoencoder is used to further reduce the feature dimensions into a uniform size of 64 for better compatibility in subsequent experiments. This target dimension was selected as a balance between reducing computational complexity and retaining the essential information from the original high-dimensional features. The intuition behind the fusion is that different VLMs focus on distinct aspects of images, and their combined features may provide a more comprehensive and richer representation of the underlying tissue structures. This fused representation was used alongside individual VLMs for comparison.

3.3 Clustering and biomarker identification

We applied the four clustering algorithms—K-Means, Agglomerative, Gaussian Mixture, and Spectral methods, to discover patterns from the features extracted by VLMs from the LungMAP H & E image patches. For each algorithm, we set the number of clusters to 7. This number was chosen as a practical and balanced choice to demonstrate the framework's capabilities. The performance of the VLMs in feature extraction is evaluated using silhouette scores, as shown in Table 2. The silhouette score measures how well each data point fits within its cluster compared to other clusters. Higher scores indicate better-defined clusters. In Fig. 4, we show a heatmap of silhouette scores for the 4-months age group to illustrate that our proposed UVLMBI approach outperforms individual VLMs across different clustering algorithms.

Table 3 summarizes the best-performing VLMs for each age group and clustering algorithm. The fused embedding-based approach achieved the highest silhouette scores for 18 out of 24 age group-cluster combinations, making it the top-performing model in most cases. However, in certain combinations, other VLMs performed better. For example, Prov-gigapath achieved the highest scores for the 4-years and 6-days groups in some cases, and Conch performed best for 9-months and 6-days age groups in specific clustering algorithms. These findings highlight the strength of using fused features for clustering. In some cases, individual VLMs can provide better-defined clusters for specific age groups and clustering methods.

The silhouette scores obtained from our experiments are generally low to moderate, with the best performance reaching a score of 0.304 for the 4-year Gaussian Mixture clustering. This reflects the inherent challenges in clustering complex and high-dimensional medical images. Such scores are not uncommon in the context of unsupervised learning on unlabeled data, particularly when dealing with noisy and heterogeneous datasets. Despite

Table 1 List of dimensions of extracted features from the patches for each VLM model

VLMs	Feature dimensions
Conch	512
UNI	1024
Prov-gigapath	1024
Llava	4096
Llava-next	4096
Plip	512
Fused	64

Table 2 Silhouette scores for all the VLMs' performances in feature extraction by clustering with cluster algorithms. Highest scores are in bold

Age Group	Cluster Algorithm	UNI	Prov-gigapath	Conch	Llava	Llava-next	Plip	Fused
1_day	K-Means	0.1468	0.153	0.1211	-0.0156	0.1526	0.0967	**0.1667**
1_day	Agglomerative	0.1718	0.1817	0.1241	0.0158	0.1186	0.1112	**0.2604**
1_day	Gaussian Mixture	0.111	0.1376	0.1232	0.0145	0.1425	0.0837	**0.175**
1_day	Spectral	0.1571	0.1319	0.1219	0.0191	0.1225	0.1023	**0.2404**
4_months	K-Means	0.1337	0.1456	0.1344	0.096	0.1271	0.1018	**0.1961**
4_months	Agglomerative	0.1387	0.1265	0.1197	0.0855	0.1156	0.0881	**0.2285**
4_months	Gaussian Mixture	0.1195	0.1261	0.1275	0.0949	0.1334	0.1067	**0.1944**
4_months	Spectral	0.1476	0.1497	0.1211	0.0135	0.1237	0.0943	**0.2843**
4_years	K-Means	0.1576	**0.192**	0.1566	0.0826	0.1195	0.1165	0.1899
4_years	Agglomerative	0.1663	**0.1966**	0.136	0.0481	0.1042	0.1238	0.1686
4_years	Gaussian Mixture	0.1438	0.1398	0.1608	0.0836	0.1223	0.1306	**0.304**
4_years	Spectral	0.1413	**0.1798**	0.1565	-0.0107	0.1206	0.1307	0.1724
6_days	K-Means	0.0962	**0.1688**	0.1047	-0.0305	0.1292	0.1016	0.1678
6_days	Agglomerative	0.1238	0.1263	0.139	0.0078	0.1376	0.1181	**0.2185**
6_days	Gaussian Mixture	0.0586	0.0953	0.1301	-0.0628	0.1088	0.106	**0.1162**
6_days	Spectral	0.1018	0.0812	0.1066	0.018	0.0911	0.1175	**0.1951**
9_months	K-Means	0.1248	0.1236	0.1088	0.0566	0.0978	0.0898	**0.145**
9_months	Agglomerative	0.1226	0.1196	0.1033	0.0371	0.093	0.0726	**0.1314**
9_months	Gaussian Mixture	0.1298	0.1263	0.0933	0.0934	0.0992	0.1002	**0.1534**
9_months	Spectral	0.1144	0.0992	**0.1217**	0.003	0.0995	0.0844	0.1039
20_months	K-Means	0.0915	0.0838	0.097	0.0196	0.0919	0.0921	**0.1094**
20_months	Agglomerative	0.1169	0.1121	0.104	-0.018	0.0771	0.0843	**0.1783**
20_months	Gaussian Mixture	0.0759	0.0596	0.0873	0.0026	0.0664	0.0814	**0.1105**
20_months	Spectral	0.0926	0.0827	0.1166	0.0165	0.0903	0.0922	**0.1235**

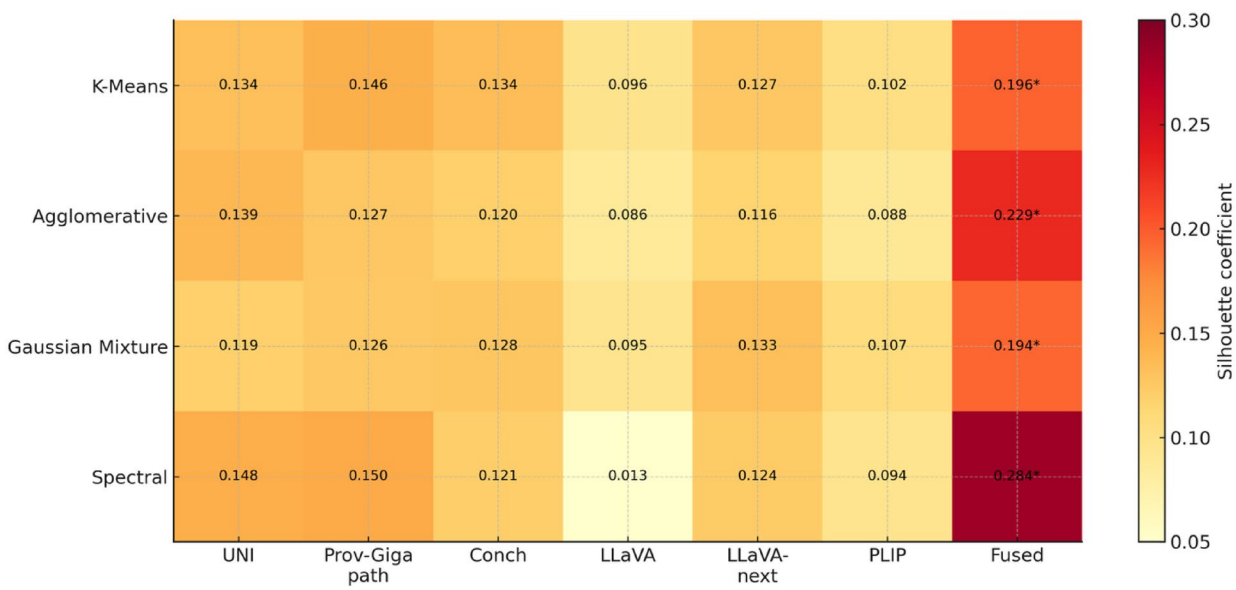

Fig. 4 Heatmap of Silhouette Scores for the 4-months Age Group. This plot visually compares the performance of each VLM and the fused approach across the four clustering algorithms. The color intensity of each cell corresponds to the silhouette score, with warmer colors indicating a higher score and better-defined clusters. This visualization highlights the superior performance of the Fused model for this age group, particularly with Spectral and Agglomerative clustering. Here y-axis shows the clustering algorithms and x-axis shows the list of different VLMs

Table 3 The best VLMs based on their silhouette scores for each Age group and Cluster algorithm combinations

Age group	Cluster algorithm	Best VLM	Silhouette score
1_day	K-Means	Fused	0.1667
1_day	Agglomerative	Fused	0.2604
1_day	Gaussian Mixture	Fused	0.175
1_day	Spectral	Fused	0.2404
4_months	K-Means	Fused	0.1961
4_months	Agglomerative	Fused	0.2285
4_months	Gaussian Mixture	Fused	0.1944
4_months	Spectral	Fused	0.2843
4_years	K-Means	Prov-gigapath	0.192
4_years	Agglomerative	Prov-gigapath	0.1966
4_years	Gaussian Mixture	Fused	0.304
4_years	Spectral	Prov-gigapath	0.1798
6_days	K-Means	Prov-gigapath	0.1678
6_days	Agglomerative	Fused	0.2185
6_days	Gaussian Mixture	Conch	0.1162
6_days	Spectral	Fused	0. 1951
9_months	K-Means	Fused	0.145
9_months	Agglomerative	Fused	0.1314
9_months	Gaussian Mixture	Fused	0.1534
9_months	Spectral	Conch	0.1217
20_months	K-Means	Fused	0.1094
20_months	Agglomerative	Fused	0.1783
20_months	Gaussian Mixture	Fused	0.1105
20_months	Spectral	Fused	0.1235

these limitations, our work demonstrates the feasibility of using fused VLM embeddings and highlights the potential for further optimization through techniques like dimensionality reduction, alternative clustering methods, or semi-supervised learning.

Figure 5 shows the 9-month age group for the fused embeddings for the fused embeddings as that provides us both early and late-stage morphological features. We try to maintain clarity and avoid overwhelming the reader with excessive combinations by focusing on limited number of age group VLM combinations. Figure 5 shows the clustering results from K-Means, Agglomerative, Gaussian Mixture, and Spectral methods for Fused embeddings for 9 months age group. While K-Means and Agglomerative clustering produced well-separated clusters, the Gaussian Mixture Model identified more granular subgroups, which could indicate smaller biomarker variations. Spectral clustering displayed less-defined separation, which might be due to the non-linearly separable nature of the data.

In Fig. 6, we show the K-Means clustering results for the 9-month age group for each VLM including the fused embeddings. The fused embeddings-based approach demonstrates more distinct and balanced clusters, which likely capture a broader range of morphological features due to the combination of embeddings from multiple models. In contrast, individual models like Llava and UNI produce clusters with less clear separation, indicating that their representations may focus on narrower aspects of the data. The fused model's ability to combine the strengths of individual VLMs results in more refined clusters, making it a more effective approach for biomarker discovery.

Figure 7 shows the ROC-AUC results for the 6-day K-Means clustering across different VLMs including the fused embeddings. Fused and Prov-Gigapath models show higher AUC scores for several classes, particularly class 3 (AUC: 0.83) and class 4 (AUC: 0.81), indicating strong predictive capabilities. The Plip model also demonstrates high performance for class 6 with an AUC of 0.82. However, models like Llava and Llava-next exhibit more moderate and consistent AUC values across most classes, reflecting a more balanced but less class-specific performance. These results suggest that different VLMs may specialize in detecting certain clusters, while the fused approach offers a comprehensive view, improving performance for specific cluster classes.

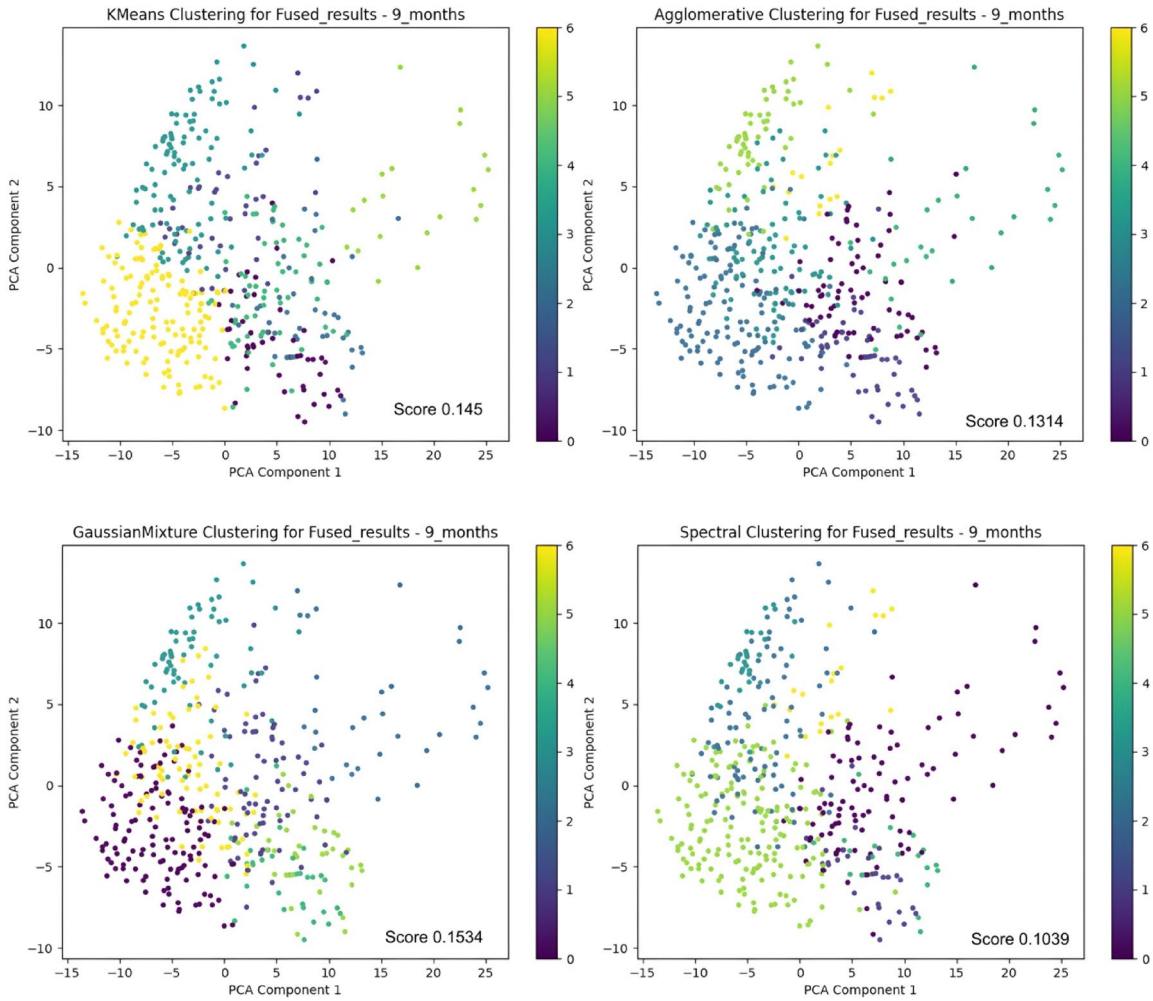

Fig. 5 Comparison of clustering algorithms (K-Means, Agglomerative, Gaussian Mixture, and Spectral) applied to the fused embeddings for the 9-month age group. The plots visualize the distinct clusters identified by each algorithm and the silhouette scores, highlighting differences in cluster separation and structure based on PCA-reduced components. The color bar represents the assigned cluster labels, with variations between methods demonstrating the impact of algorithm selection on biomarker identification

Figure 8 shows the ROC-AUC curves for the 9-month K-Means clustering results for different VLMs including the fused embeddings. Models like Llava-next and Prov-Gigapath show higher AUC values, particularly for classes 0 and 6, indicating strong predictive performance for certain clusters. UNI shows the highest AUC (0.72) for class 2, demonstrating its strength in capturing specific clusters with accuracy. Conch and fused exhibit more balanced yet lower AUC values across all classes, reflecting a more generalized but less discriminative performance. These comparisons suggest that while individual models may excel in identifying specific clusters, the fused approach offers a broader view of the overall data structure, albeit with slightly lower class-specific precision.

3.4 Biomarker prediction using metadata

Once the clusters were formed, we predicted relevant biomarkers by integrating clinical metadata. This experiment aimed link image-derived clusters with the real-world clinical metadata, such as patients' demographics, clinical history, and molecular data. Our experiments with various machine learning classifiers shows that random

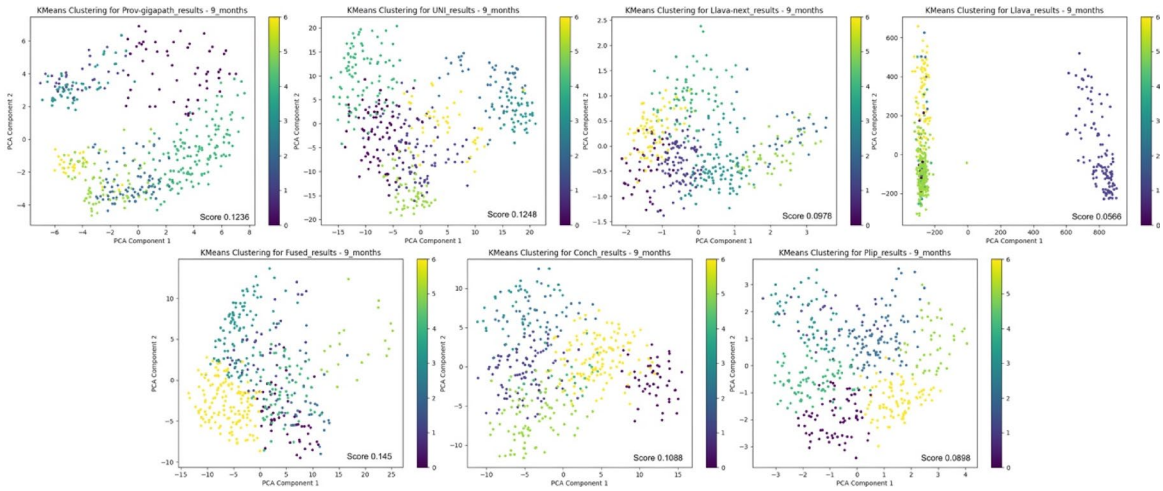

Fig. 6 Comparison of VLMs with K-means clustering applied to the fused embeddings for the 9-month age group. The plots visualize the distinct clusters identified by each algorithm and the silhouette scores, highlighting differences in cluster separation and structure based on PCA-reduced components. The color bar represents the assigned cluster labels, with variations between methods demonstrating the impact of algorithm selection on biomarker identification

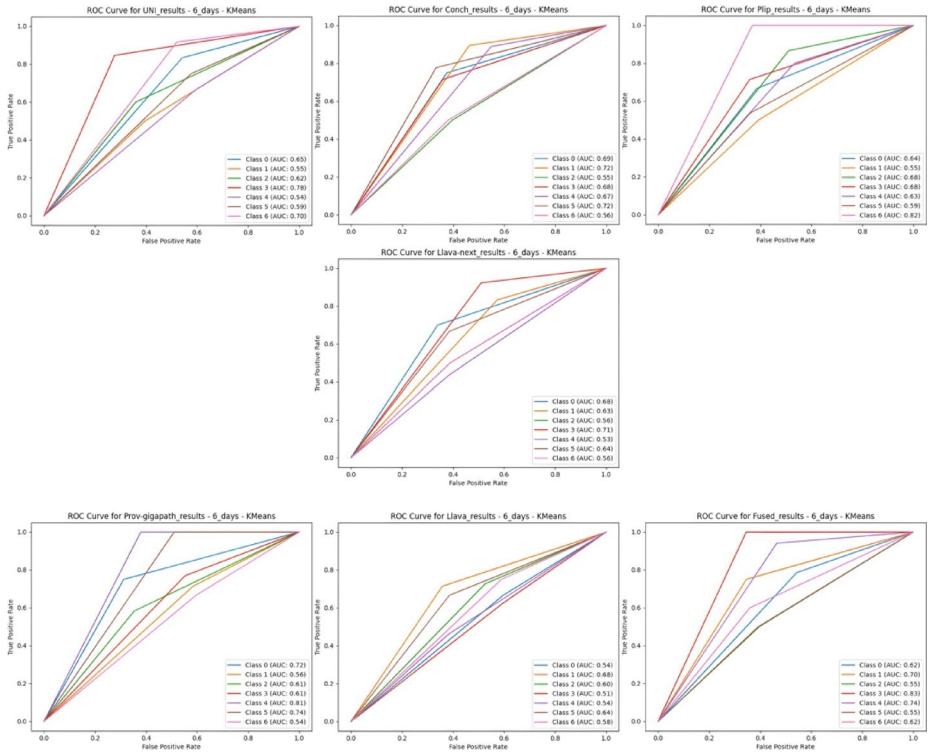

Fig. 7 ROC-AUC curves for the 6-day age group using K-Means clustering across different VLMs, namely UNI, Conch, Plip, Llava-next, Prov-Gigapath, Llava, and Fused approach. Each curve illustrates the trade-off between true positive and false positive rates for various classes, providing insights into the model performance in classifying clusters. Here, classes 0–6 refer to the seven clusters generated by the K-Means algorithm. These ROC-AUC curves demonstrate the classifier's performance in predicting each cluster from the clinical metadata

forest classifier performed well. Hence, for each age group, a random forest classifier was trained on the metadata to predict the biomarker clusters formed by the VLM embeddings. The metadata features used primarily for the biomarker prediction included donor age, sex, body weight, magnification level, and cause of death etc. Random forest classifier was used with hyperparameter tuning performed via Optuna, an optimization framework, to identify the best combination of parameters. The final classifier was configured with the number of estimators set to 100 (number of trees), a maximum tree depth of 10, a minimum sample size of 2 for splitting, and 1 for leaf

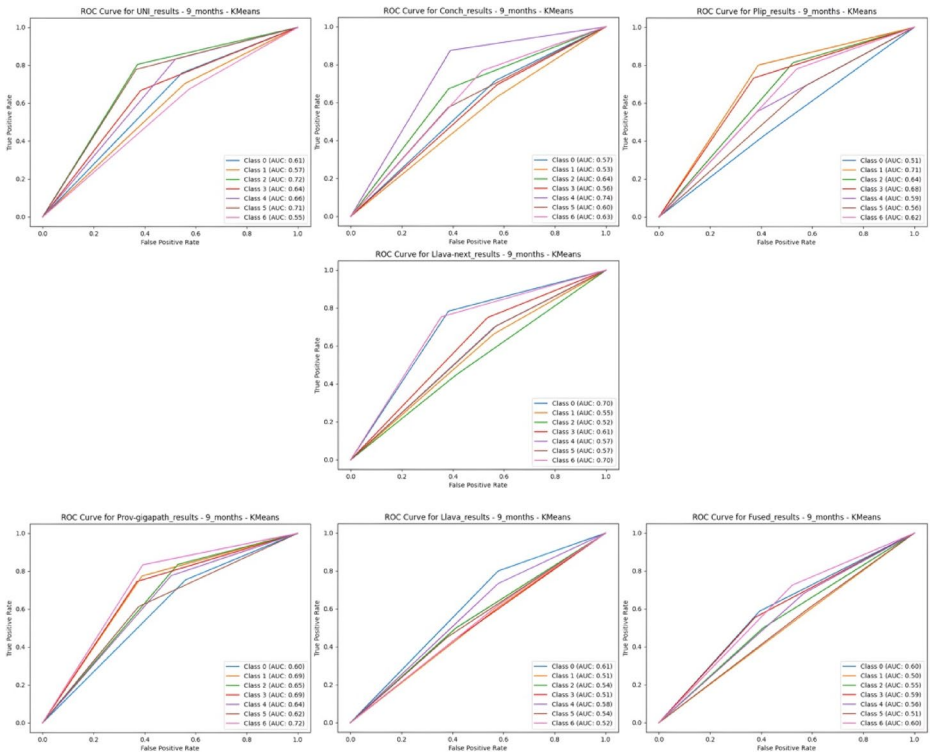

Fig. 8 ROC-AUC curves for the 9-month age group using K-Means clustering across different VLMs, namely UNI, Conch, Plip, Llava-next, Prov-Gigapath, Llava, and Fused approach. Each curve represents the true positive rate (sensitivity) vs. the false positive rate (1-specificity) for different classes, illustrating the models' performance in distinguishing between various clusters. Here, classes 0–6 correspond to the seven clusters identified by the K-Means algorithm, with each class representing a potential biomarker. This figure compares the predictive performance of these clusters across different VLMs

nodes. We used ROC-AUC scores to measure the VLM performances, as it provides a robust measure of classifier performance in the presence of class imbalances, which are common in medical datasets.

Table 4 shows the results of biomarkers prediction for 24 combinations (age group and cluster algorithms) for each VLM and fused model. From this we notice that for 4 combinations (1 day – K-Means, 4 years—Agglomerative, 4 years – Spectral and 6 days – K-Means) the ROC-AUC scores for the fused model were the highest and for other 4 combinations (4 months – Gaussian Mixture, 4 years – K-Means, 9 months – Spectral, and 20 months – Gaussian Mixture) the fused model performed the second highest. This result demonstrates the effectiveness of the fusion model in capturing diverse feature representations from multiple VLMs, which is particularly useful when analyzing biological data across varied age groups and clustering techniques. However, we note that, the reported results do not convincingly show significant performance improvements compared to using individual models. This is interesting, since we believe fusion of multiple VLMs may introduce additional noise and bias, rather than enhancing model performance.

Table 5 shows the best VLM performances for each age group – cluster algorithm combinations. UNI was the top performer in 7 out of 24 combinations. Though the fused was best in 4 out of 24 combinations, this is notable that other individual VLMs often focus on specific image features, limiting their generalizability across different contexts. Even though the fused model was not the best for most combinations, its strong performance in highly diverse clustering techniques and age groups highlights its flexibility and potential in biomarker discovery, especially in an unsupervised setting. As can be seen from Table 5, for 1 day age group Llava VLM obtained the best results overall, with GMM obtained a highest ROC AUC score of 0.75. For 20 months age group, Llava VLM obtained the highest ROC AUC score of 0.76 with K-means clustering. For 4 months age group, Llava and UNI VLMs performed on par with ROC AUC score of 0.54. For 4 years age group fused was the best with ROC AUC 0.64 and UNI with ROC AUC of 0.60. For 6 days age group, Prov-gigapath obtained best ROC AUC 0.70 with Llava-next the next best with 0.67. For 9 months age group, both UNI and Prov-gigapath performed equally well with ROC AUC score of 0.68. Notably, fused outperformed other VLMs in the 1-day, 6-day and 4-years age groups across several clustering algorithms, demonstrating its strength in these biological contexts. Furthermore,

Table 4 Overall performance for all the VLMs and fused for all the age group—cluster algorithm combinations

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.58
1_day	Agglomerative	0.58
1_day	Gaussian Mixture	0.57
1_day	Spectral	0.55
4_months	K-Means	0.54
4_months	Agglomerative	0.51
4_months	Gaussian Mixture	0.52
4_months	Spectral	0.52
4_years	K-Means	0.57
4_years	Agglomerative	0.58
4_years	Gaussian Mixture	0.57
4_years	Spectral	0.58
6_days	K-Means	0.65
6_days	Agglomerative	0.62
6_days	Gaussian Mixture	0.66
6_days	Spectral	0.62
9_months	K-Means	0.61
9_months	Agglomerative	0.61
9_months	Gaussian Mixture	0.57
9_months	Spectral	0.62
20_months	K-Means	0.64
20_months	Agglomerative	0.65
20_months	Gaussian Mixture	0.67
20_months	Spectral	0.62

(a) Conch

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.59
1_day	Agglomerative	0.63
1_day	Gaussian Mixture	0.75
1_day	Spectral	0.64
4_months	K-Means	0.55
4_months	Agglomerative	0.53
4_months	Gaussian Mixture	0.52
4_months	Spectral	0.52
4_years	K-Means	0.53
4_years	Agglomerative	0.53
4_years	Gaussian Mixture	0.55
4_years	Spectral	0.55
6_days	K-Means	0.58
6_days	Agglomerative	0.57
6_days	Gaussian Mixture	0.61
6_days	Spectral	0.57
9_months	K-Means	0.55
9_months	Agglomerative	0.53
9_months	Gaussian Mixture	0.52
9_months	Spectral	0.56
20_months	K-Means	0.76
20_months	Agglomerative	0.63
20_months	Gaussian Mixture	0.63
20_months	Spectral	0.64

(b) Llava

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.56
1_day	Agglomerative	0.61
1_day	Gaussian Mixture	0.61
1_day	Spectral	0.59
4_months	K-Means	0.54
4_months	Agglomerative	0.53
4_months	Gaussian Mixture	0.53
4_months	Spectral	0.53
4_years	K-Means	0.59
4_years	Agglomerative	0.58
4_years	Gaussian Mixture	0.6
4_years	Spectral	0.58
6_days	K-Means	0.61
6_days	Agglomerative	0.66
6_days	Gaussian Mixture	0.62
6_days	Spectral	0.67
9_months	K-Means	0.6
9_months	Agglomerative	0.6
9_months	Gaussian Mixture	0.6
9_months	Spectral	0.62
20_months	K-Means	0.67
20_months	Agglomerative	0.68
20_months	Gaussian Mixture	0.68
20_months	Spectral	0.66

(c) Llava-Next

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.57
1_day	Agglomerative	0.59
1_day	Gaussian Mixture	0.58
1_day	Spectral	0.59
4_months	K-Means	0.53
4_months	Agglomerative	0.54
4_months	Gaussian Mixture	0.53
4_months	Spectral	0.52
4_years	K-Means	0.54
4_years	Agglomerative	0.56
4_years	Gaussian Mixture	0.54
4_years	Spectral	0.62
6_days	K-Means	0.66
6_days	Agglomerative	0.63
6_days	Gaussian Mixture	0.64
6_days	Spectral	0.63
9_months	K-Means	0.61
9_months	Agglomerative	0.62
9_months	Gaussian Mixture	0.62
9_months	Spectral	0.64
20_months	K-Means	0.63
20_months	Agglomerative	0.63
20_months	Gaussian Mixture	0.63
20_months	Spectral	0.65

(d) Plip

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.59
1_day	Agglomerative	0.57
1_day	Gaussian Mixture	0.59
1_day	Spectral	0.58
4_months	K-Means	0.52
4_months	Agglomerative	0.51
4_months	Gaussian Mixture	0.52
4_months	Spectral	0.52
4_years	K-Means	0.6
4_years	Agglomerative	0.61
4_years	Gaussian Mixture	0.56
4_years	Spectral	0.61
6_days	K-Means	0.66
6_days	Agglomerative	0.66
6_days	Gaussian Mixture	0.7
6_days	Spectral	0.66
9_months	K-Means	0.66
9_months	Agglomerative	0.62
9_months	Gaussian Mixture	0.63
9_months	Spectral	0.68
20_months	K-Means	0.71
20_months	Agglomerative	0.7
20_months	Gaussian Mixture	0.72
20_months	Spectral	0.73

(e) Prov-gigapath

Age Group	Algorithm	ROC AUC
1_day	K-Means	0.57
1_day	Agglomerative	0.57
1_day	Gaussian Mixture	0.6
1_day	Spectral	0.57
4_months	K-Means	0.53
4_months	Agglomerative	0.54
4_months	Gaussian Mixture	0.52
4_months	Spectral	0.55
4_years	K-Means	0.6
4_years	Agglomerative	0.62
4_years	Gaussian Mixture	0.6
4_years	Spectral	0.63
6_days	K-Means	0.63
6_days	Agglomerative	0.65
6_days	Gaussian Mixture	0.65
6_days	Spectral	0.66
9_months	K-Means	0.64
9_months	Agglomerative	0.68
9_months	Gaussian Mixture	0.63
9_months	Spectral	0.64
20_months	K-Means	0.69
20_months	Agglomerative	0.71
20_months	Gaussian Mixture	0.66
20_months	Spectral	0.72

(f) UNI

Age Group	Algorithm	RC AU
1_day	K-Means	0.6
1_day	Agglomerative	0.59
1_day	Gaussian Mixture	0.6
1_day	Spectral	0.57
4_months	K-Means	0.53
4_months	Agglomerative	0.53
4_months	Gaussian Mixture	0.52
4_months	Spectral	0.52
4_years	K-Means	0.6
4_years	Agglomerative	0.62
4_years	Gaussian Mixture	0.58
4_years	Spectral	0.64
6_days	K-Means	0.66
6_days	Agglomerative	0.66
6_days	Gaussian Mixture	0.66
6_days	Spectral	0.65
9_months	K-Means	0.56
9_months	Agglomerative	0.62
9_months	Gaussian Mixture	0.61
9_months	Spectral	0.64
20_months	K-Means	0.7
20_months	Agglomerative	0.66
20_months	Gaussian Mixture	0.69
20_months	Spectral	0.64

(g) Fused

the overall stability of fused across multiple conditions indicates that it offers robust performance, making it a versatile choice for biomarker discovery.

Figure 9 shows the ROC–AUC scores across age groups and clustering algorithms in a heatmap visualization. We note the performance of the best-performing VLMs for biomarker prediction across different age groups and clustering methods, highlighting consistent high-AUC regions across models like UNI, Prov-Gigapath, and Llava. This underscors their robustness in biomarker classification across age progression.

Table 5 The best VLM performance for each age group—cluster algorithm combinations. Highest scores are in **bold**

VLM	Age group	Algorithm	ROC AUC
Llava_results	1_day	Agglomerative	0.63
Llava_results	1_day	Gaussian Mixture	**0.75**
Fused_results	1_day	K-Means	0.60
Llava_results	1_day	Spectral	0.64
UNI_results	20_months	Agglomerative	0.71
Prov-gigapath_results	20_months	Gaussian Mixture	0.72
Llava_results	20_months	K-Means	**0.76**
Prov-gigapath_results	20_months	Spectral	0.73
UNI_results	4_months	Agglomerative	0.54
Plip_results	4_months	Gaussian Mixture	0.53
Llava_results	4_months	K-Means	**0.55**
UNI_results	4_months	Spectral	0.55
Fused_results	4_years	Agglomerative	0.62
UNI_results	4_years	Gaussian Mixture	0.60
UNI_results	4_years	K-Means	0.60
Fused_results	4_years	Spectral	**0.64**
Llava-next_results	6_days	Agglomerative	0.66
Prov-gigapath_results	6_days	Gaussian Mixture	**0.70**
Fused_results	6_days	K-Means	0.66
Llava-next_results	6_days	Spectral	0.67
UNI_results	9_months	Agglomerative	**0.68**
UNI_results	9_months	Gaussian Mixture	0.63
Prov-gigapath_results	9_months	K-Means	0.66
Prov-gigapath_results	9_months	Spectral	**0.68**

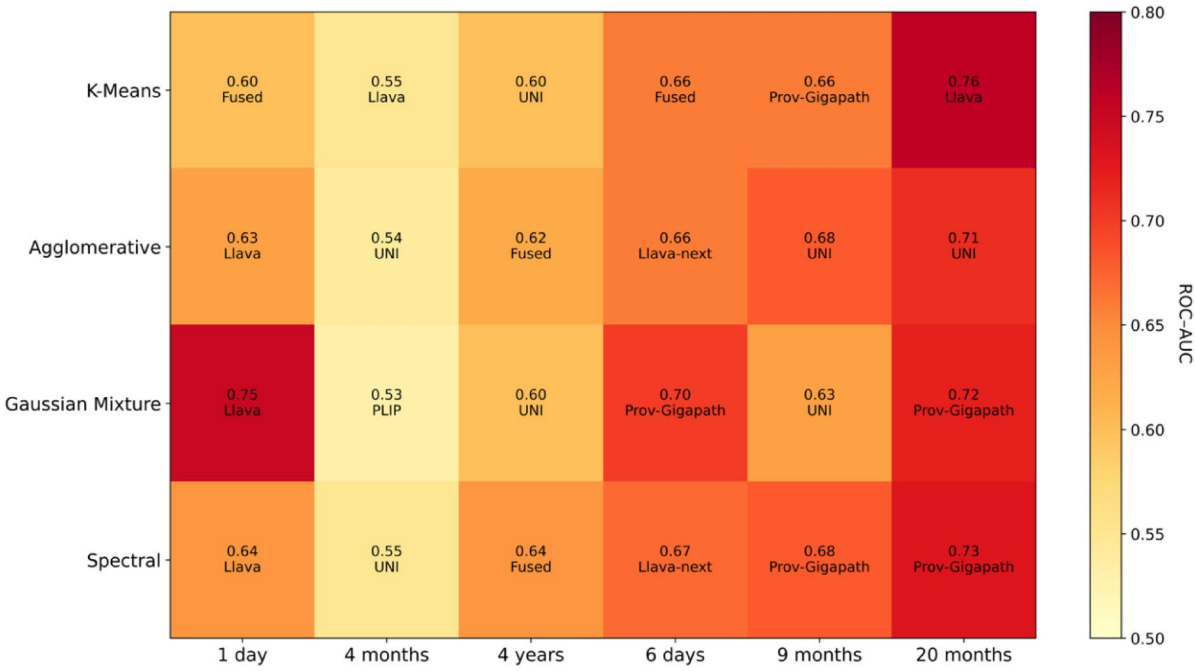

Fig. 9 Heatmap of ROC–AUC Scores across Age Groups and Clustering Algorithms. This plot presents the performance of the best-performing VLMs for biomarker prediction across different age groups and clustering methods. Each cell also denotes the VLM achieving the best score for that specific age–algorithm combination. This visualization highlights consistent high-AUC regions across models like UNI, Prov-Gigapath, and Llava, underscoring their robustness in biomarker classification across age progression

4 Discussion

The experimental results highlight some key findings using our proposed UVLMBI framework. First, the effectiveness of the VLMs in biomarker discovery is influenced by both the clustering algorithm and metadata of the tissue samples. Fused VLMs embeddings, though not consistently superior, demonstrated enhanced performances in some combinations. This suggests that the fused model can capture some additional effective features which other individual models might miss. Second, metadata integration plays an important role in making the clusters interpretable and clinically relevant. By linking clusters to patient demographics and clinical history, the UVLMBI framework provides a path toward personalized medicine, where the biomarkers are not only discovered, but also contextualized within patients' clinical background. Finally hyper parameter tuning and effective ensembling processes among different machine learning classifiers may improve biomarker prediction performance. The high silhouette scores observed with fused embeddings indicate that this method captures a more comprehensive representation of histological features compared to individual VLMs. This fusion methodology can be extended further in multimodal domains, such as spatial transcriptomics and RNASeq, to understand the tissue morphological microenvironment. In this paper, we evaluated four histopathology specific VLMs and two generic image-text VLMs to analyze the strength of specialized histopathology models alongside the flexibility of general-purpose models in unsupervised biomarker finding scenario. While models such as Llava-Med [26] (trained with radiology images like MRI, CT scans, and histology images) and Quilt-Llava [27] (trained on histopathology videos) represent valuable alternatives, they were not included in this study due to their broader focus beyond histopathology WSIs. However, future experiments can explore these models image encoders along with other relevant model options, such as BiomedCLIP [20] or MedCLIP [28]. These models are trained on diverse medical datasets and can be explored to assess whether they further enhance clustering and biomarker prediction.

Future work could explore the integration of different datasets of different modalities, like single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics [29], to enhance the depth of biomarker discovery. The multi-omics approach can analyze the combination of imaging and molecular data which uncovers complex biological relationships that are possibly missed in unimodal analyses. Another promising direction of further enhancement is to introduce advanced neural network architectures, such as graph neural networks (GNNs) [30], which could better capture spatial relationships in tissue structures. Additionally, the incorporation of larger and more diverse datasets from multiple organs and sources can improve the generalizability and robustness of the UVLMBI framework. Transfer learning techniques could also help adapt this framework to new disease contexts with minimal retraining, making it more applicable in personalized medicine and clinical decision-making. Future research can focus on refining the clustering techniques by experimenting with non-linear dimensionality reduction methods, such as t-SNE [31] or UMAP [32], to reduce noise in the high-dimensional embeddings. Additionally, incorporating other clinical metadata more effectively or using graph-based clustering methods could yield more meaningful biomarker groupings. A semi-supervised learning approach, leveraging partially labeled data, may also improve cluster separability. Extending the methodology to other datasets, including spatial transcriptomics, could enhance the generalizability and applicability of the framework. Another potential research path can be in exploring not only the integration of additional omics data types but also the optimization of the 'Fused' approach by considering the combination of only the top 2 or 3 best-performing VLMs to reduce computational overhead and potentially enhance overall performance. These advancements aim to further enhance biomarker discovery and validation by creating a more efficient and powerful pipeline.

5 Conclusion

The UVLMBI framework successfully demonstrated the potential of combining vision language models and clinical metadata in unsupervised biomarker discovery. The testing of multiple vision language models (VLMs) and their embeddings provided a more comprehensive understanding of tissue features across different age groups.

This approach not only identified potential biomarkers but also linked them to clinically relevant metadata, opening new possibilities for personalized treatment strategies. While the framework shows promising results, further work is needed to optimize the fusion process and biomarker prediction process. Integration of additional modalities can be a good possible approach to this. The UVLMBI framework could be a cornerstone for future research in digital pathology, particularly in enhancing the precision and personalization of medical treatments. While the clustering results achieved in this study show promising performance, they provide a solid foundation for future research. Improvements such as incorporating more advanced dimensionality reduction techniques, exploring additional clustering algorithms, and integrating larger, labeled datasets in a semi-supervised manner could yield better-defined clusters and improved biomarker discovery. This study demonstrates the feasibility of using fused vision-language models and serves as an exploratory framework for future enhancements in the field of digital pathology.

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Data availability The data will be made available on reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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Consent to participate Not applicable.

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References

1. A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin and J. Clark, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning,* 2021, pp. 8748–8763.
2. C. Jia, Y. Yang, Y. Xia, Y. Chen, Z. Parekh, H. Pham, Q. Le, Y. Sung, Z. Li and T. Duerig, "Scaling up visual and vision-language representation learning with noisy text supervision," in *International Conference on Machine Learning,* 2021, pp. 4904–4916.
3. Alroobaea R, Rubaiee S, Bourouis S, Bouguila N, Alsufyani A (2020) Bayesian inference framework for bounded generalized Gaussian-based mixture model and its application to biomedical images classification. Int J Imaging Syst Technol 30(1):18–30
4. Ortega S, Halicek M, Fabelo H, Callico GM, Fei B (2020) Hyperspectral and multispectral imaging in digital and computational pathology: a systematic review. Biomed Opt Express 11(6):3195–3233

5. Diao JA, Wang JK, Chui WF, Mountain V, Gullapally SC, Srinivasan R, Mitchell RN, Glass B, Hoffman S, Rao SK (2021) Human-interpretable image features derived from densely mapped cancer pathology slides predict diverse molecular phenotypes. Nat Commun 12(1):1613
6. Lee J, Warner E, Shaikhouni S, Bitzer M, Kretzler M, Gipson D, Pennathur S, Bellovich K, Bhat Z, Gadegbeku C (2022) Unsupervised machine learning for identifying important visual features through bag-of-words using histopathology data from chronic kidney disease. Sci Rep 12(1):4832
7. Yang Y, Yu J, Zhang J, Han W, Jiang H, Huang Q (2021) Joint embedding of deep visual and semantic features for medical image report generation. IEEE Trans Multimedia 25:167–178
8. Dai Y, Gao Y, Liu F (2021) Transmed: transformers advance multi-modal medical image classification. Diagnostics 11(8):1384
9. Huo X, Sun G, Tian S, Wang Y, Yu L, Long J, Zhang W, Li A (2024) HiFuse: hierarchical multi-scale feature fusion network for medical image classification. Biomed Signal Process Control 87:105534
10. Steyaert S, Pizurica M, Nagaraj D, Khandelwal P, Hernandez-Boussard T, Gentles AJ, Gevaert O (2023) Multimodal data fusion for cancer biomarker discovery with deep learning. Nat Mach Intell 5(4):351–362
11. Stahlschmidt SR, Ulfennborg B, Synnergren J (2022) Multimodal deep learning for biomedical data fusion: a review. Brief Bioinform 23(2):bbab569
12. N. Braman, J. W. Gordon, E. T. Goossens, C. Willis, M. C. Stumpe and J. Venkataraman, "Deep orthogonal fusion: Multimodal prognostic biomarker discovery integrating radiology, pathology, genomic, and clinical data," in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part V 24,* 2021, pp. 667–677.
13. Gaddis N, Fortriede J, Guo M, Bardes EE, Kouril M, Tabar S, Burns K, Ardini-Poleske ME, Loos S, Schnell D (2024) LungMAP portal ecosystem: systems-level exploration of the lung. Am J Respir Cell Mol Biol 70(2):129–139
14. Chen RJ, Ding T, Lu MY, Williamson DF, Jaume G, Song AH, Chen B, Zhang A, Shao D, Shaban M (2024) Towards a general-purpose foundation model for computational pathology. Nat Med 30(3):850–862
15. Xu H, Usuyama N, Bagga J, Zhang S, Rao R, Naumann T, Wong C, Gero Z, González J, Gu Y (2024) A whole-slide foundation model for digital pathology from real-world data. Nature. https://doi.org/10.1038/s41586-024-07441-w
16. Lu MY, Chen B, Williamson DF, Chen RJ, Liang I, Ding T, Jaume G, Odintsov I, Le LP, Gerber G (2024) A visual-language foundation model for computational pathology. Nat Med 30(3):863–874
17. H. Liu, C. Li, Q. Wu and Y. J. Lee, "Visual instruction tuning," *Advances in Neural Information Processing Systems,* vol. 36, 2024.
18. Liu, Haotian and Li, Chunyuan and Li, Yuheng and Li, Bo and Zhang, Yuanhan and Shen, Sheng and Lee, Yong Jae, "LLaVA-NeXT: Improved reasoning, OCR, and world knowledge," 2024. Available: https://llava-vl.github.io/blog/2024-01-30-llava-next/.
19. Huang Z, Bianchi F, Yuksekgonul M, Montine TJ, Zou J (2023) A visual–language foundation model for pathology image analysis using medical twitter. Nat Med 29(9):2307–2316
20. S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei and N. Valluri, "BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," *arXiv Preprint* arXiv:2303.00915*,* 2023. .
21. Zheng L, Chiang W, Sheng Y, Zhuang S, Wu Z, Zhuang Y, Lin Z, Li Z, Li D, Xing E (2023) Judging llm-as-a-judge with mt-bench and chatbot arena. Adv Neural Inf Process Syst 36:46595–46623
22. G. Team, R. Anil, S. Borgeaud, Y. Wu, J. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai and A. Hauth, "Gemini: a family of highly capable multimodal models," *arXiv Preprint* arXiv:2312.11805*,* 2023.
23. J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou and J. Zhou, "Qwen-vl: A frontier large vision-language model with versatile abilities," *arXiv Preprint* arXiv:2308.12966*,* 2023.
24. Wang X, Yang S, Zhang J, Wang M, Zhang J, Yang W, Huang J, Han X (2022) Transformer-based unsupervised contrastive learning for histopathological image classification. Med Image Anal 81:102559
25. S. Azizi, L. Culp, J. Freyberg, B. Mustafa, S. Baur, S. Kornblith, T. Chen, P. MacWilliams, S. S. Mahdavi and E. Wulczyn, "Robust and efficient medical imaging with self-supervision," *arXiv Preprint* arXiv:2205.09723*,* 2022.
26. C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon and J. Gao, "Llava-med: Training a large language-and-vision assistant for biomedicine in one day," *Advances in Neural Information Processing Systems,* vol. 36, 2024.
27. M. S. Seyfioglu, W. O. Ikezogwo, F. Ghezloo, R. Krishna and L. Shapiro, "Quilt-LLaVA: Visual instruction tuning by extracting localized narratives from open-source histopathology videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 2024, pp. 13183–13192.
28. Z. Wang, Z. Wu, D. Agarwal and J. Sun, "Medclip: Contrastive learning from unpaired medical images and text," *arXiv Preprint* arXiv:2210.10163*,* 2022.
29. Rajdeo P, Aronow B, Prasath VS (2024) Deep learning-based multimodal spatial transcriptomics analysis for cancer. Adv Cancer Res 163:1–38

30. Wu Z, Pan S, Chen F, Long G, Zhang C, Philip SY (2020) A comprehensive survey on graph neural networks. IEEE Trans Neural Netw Learn Syst 32(1):4–24
31. L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE." *Journal of Machine Learning Research,* vol. 9, *(11),* 2008.
32. L. McInnes, J. Healy and J. Melville, "Umap: Uniform manifold approximation and projection for dimension reduction," *arXiv Preprint* arXiv:1802.03426*,* 2018.
33. Ding, Jiayu, et al. "Longnet: Scaling transformers to 1,000,000,000 tokens." *arXiv preprint* arXiv:2307.02486 (2023).

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